

ConvexOptimization 1

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PROBLEM I A set C is midpoint convex if whenever $a, b \in C$, the average or midpoint $\frac{a+b}{2} \in C$. Obviously a convex set is midpoint convex. It can be proved that under mild conditions midpoint convexity implies convexity. As a simple case, prove that if C is closed and midpoint convex, then C is convex.

SOLUTION. Need to prove $\forall a, b \in C, \forall \lambda \in (0, 1), \lambda a + (1 - \lambda)b \in C$. For convinence, write $f(\lambda) = \lambda a + (1 - \lambda)b$.

First, we prove $f(\lambda) \in C$ for $\lambda = \frac{k}{2^n}$ for $k = 0, 1, 2, \dots, 2^n$.

We prove it by induction. For $n = 1$, we have $k = 0, 1, 2$. For $k = 0, 2$, obvious. For $k = 1$, it's the definition of midpoint convex.

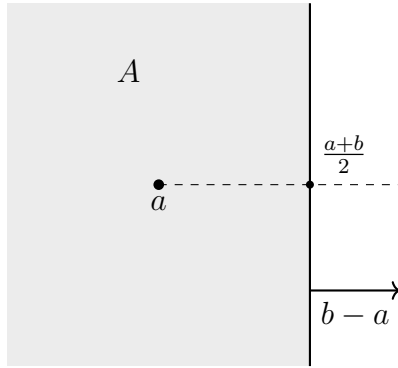
Assume for some $n = t \geq 1$, we've proved $f(\frac{k}{2^t}) \in C$, where $k = 0, \dots, 2^t$. Now consider $\lambda = \frac{k}{2^{t+1}}, k = 0, \dots, 2^{t+1}$. If $2 \mid k$, then $k = 2m$, then $\lambda = \frac{m}{2^t}, 0 \leq m \leq 2^t$. So by induction assumption $f(\lambda) \in C$. If $2 \nmid k$, then $2 \mid k - 1, k + 1$, and $0 \leq k - 1 < k + 1 \leq 2^{t+1}$. By induction assumption, we have $f(\frac{k-1}{2^{t+1}}) \in C, f(\frac{k+1}{2^{t+1}}) \in C$. Then $f(\frac{k}{2^{t+1}}) = \frac{1}{2}f(\frac{k-1}{2^{t+1}}) + \frac{1}{2}f(\frac{k+1}{2^{t+1}})$. By midpoint convex we have $f(\frac{k}{2^{t+1}}) \in C$.

By induction we have $f(\lambda) \in C$ for $\lambda = \frac{k}{2^n}, n \geq 1, 0 \leq k \leq 2^n$.

Since C is closed, for any $\lambda \in (0, 1)$, we can define $\lambda_n = \frac{\lfloor \lambda 2^n \rfloor}{2^n} \rightarrow \lambda$, and $f(\lambda_n) \in C$, and obviously f is continuous. So $f(\lambda) \in C$. So C is convex. $\square$

PROBLEM II $a, b \in \mathbb{R}^n, a \neq b$. Show $A := \{x \mid \|x - a\|_2 \leq \|x - b\|_2\}$ is a halfspace. Describe it explicitly as an inequality of the form $c^T x \leq d$. Draw a picture.

SOLUTION. $\|x - a\|^2 = (x - a)^T(x - a) \leq (x - b)^T(x - b) = \|x - b\|^2$, then rewrite the middle part, we get $(2b - 2a)^T x \leq \|b\|^2 - \|a\|^2$. Notice that $\frac{\|b\|^2 - \|a\|^2}{2} = (b - a)^T \frac{a+b}{2}$. Hence the boundary hyperplane is $(b - a)^T (x - \frac{a+b}{2}) = 0$. It passes through the midpoint of a and b , and its normal vector is $b - a$. Therefore it is the perpendicular bisector of the line segment joining a and b . The halfspace A is the side containing a .



□

PROBLEM III Which of the following sets S are polyhedra? If possible, express S in the form $S = \{x \mid Ax \preceq b, Fx = g\}$.

1. $S = \{y_1 a_1 + y_2 a_2 \mid -1 \leq y_1 \leq 1, -1 \leq y_2 \leq 1\}$, where $a_1, a_2 \in \mathbb{R}^n$.
2. $S = \{x \in \mathbb{R}^n \mid x \succeq 0, 1^T x = 1, \sum_{i=1}^n x_i a_i = b_1, \sum_{i=1}^n x_i a_i^2 = b_2\}$, where $a_i, b_j \in \mathbb{R}, 1 \leq i \leq n, j = 1, 2$.
3. $S = \{x \in \mathbb{R}^n \mid x \succeq 0, x^T y \leq 1, \forall y : \|y\|_2 = 1\}$.
4. $S = \{x \in \mathbb{R}^n \mid x \succeq 0, x^T y \leq 1 \forall y : \sum_{i=1}^n |y_i| = 1\}$.

SOLUTION. 1. Let

$$M = \begin{bmatrix} a_1 & a_2 \end{bmatrix}.$$

Then

$$S = \{My \mid -\mathbf{1} \preceq y \preceq \mathbf{1}\}.$$

Thus S is the linear image of the box $[-1, 1]^2$. Equivalently,

$$S = \text{conv}\{a_1 + a_2, a_1 - a_2, -a_1 + a_2, -a_1 - a_2\}.$$

Hence S is a polytope, and therefore a polyhedron.

If a_1, a_2 are linearly independent, define

$$L = (M^T M)^{-1} M^T.$$

Then

$$S = \{x \mid -\mathbf{1} \preceq Lx \preceq \mathbf{1}, (I - ML)x = 0\}.$$

2. All the constraints are linear in x . Define

$$a = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}, \quad a^{(2)} = \begin{bmatrix} a_1^2 \\ \vdots \\ a_n^2 \end{bmatrix}.$$

Then

$$S = \left\{ x \mid -Ix \preceq 0, \begin{bmatrix} \mathbf{1}^T \\ a^T \\ (a^{(2)})^T \end{bmatrix} x = \begin{bmatrix} 1 \\ b_1 \\ b_2 \end{bmatrix} \right\}.$$

Hence S is a polyhedron.

3. We have

$$x^T y \leq 1 \quad \text{for all } y \text{ with } \|y\|_2 = 1$$

if and only if

$$\sup_{\|y\|_2=1} x^T y \leq 1.$$

By the Cauchy–Schwarz inequality,

$$\sup_{\|y\|_2=1} x^T y = \|x\|_2.$$

Hence

$$S = \{x \mid x \succeq 0, \|x\|_2 \leq 1\}.$$

For $n \geq 2$, this set has a curved spherical boundary and therefore cannot be represented by finitely many linear inequalities. Thus it is not a polyhedron.

4. The condition

$$\sum_{i=1}^n |y_i| = 1$$

is equivalent to $\|y\|_1 = 1$. Hence

$$\sup_{\|y\|_1=1} x^T y = \|x\|_\infty.$$

Therefore

$$S = \{x \mid x \succeq 0, \|x\|_\infty \leq 1\} = \{x \mid 0 \leq x_i \leq 1, i = 1, \dots, n\}.$$

Thus S is a polyhedron, with

$$\begin{bmatrix} -I \\ I \end{bmatrix} x \preceq \begin{bmatrix} 0 \\ \mathbf{1} \end{bmatrix}.$$

□

PROBLEM IV Finish the proof of the separating hyperlane theorem: Show that a separating hyperlane exists for C, D , where C, D are convex and $C \cap D = \emptyset$.

SOLUTION. Consider $S = \{x - y : x \in C, y \in D\}$, which is convex due to $\lambda(x_1 - y_1) + (1 - \lambda)(x_2 - y_2) = (\lambda x_1 + (1 - \lambda)x_2) - (\lambda y_1 + (1 - \lambda)y_2) \in S$. We only need to prove $\exists a$ satisfies $\{a^T z \leq 0 \mid z \in S\}$. Since $\forall z \in S$, if $a^T z \leq 0$, then $\forall x \in C, y \in D$, $a^T(x - y) \leq 0$, so $a^T x \leq a^T y$. So $\sup_{x \in C} a^T x \leq \inf_{y \in D} a^T y$. Let u satisfy $\sup_{x \in C} a^T x \leq u \leq \inf_{y \in D} a^T y$. Thus, $a^T x \leq u, \forall x \in C, a^T y \geq u, \forall y \in D$.

Let $d = \inf_{s \in S} \|s\|$. If $d > 0$, then $d = \inf_{s \in S} \|s\| = \inf_{x \in C, y \in D} \|x - y\| > 0$, we can find separating hyperlane to separate S and $\{0\}$, which is solved by 2.5.1 in textbook. So we only need

to consider $d = 0$. Since $d = \inf_{s \in S} \|s\|$, we can get $0 \in \overline{S}$, where $\overline{S}$ is the closure of S . Another thing $C \cap D = \emptyset$, then $0 \notin S$, thus $0 \in S^c$. Since $0 \in \partial S$, where ∂S is the boundary of S , we can get $0 \in \partial S^c$. So $\exists t_k \in \partial S^c, k \in \mathbb{N}$ satisfies $t_k \rightarrow 0, k \rightarrow \infty$. Let $p_k = \arg \min_{p \in \overline{S}} \|t_k - p\|, k \in \mathbb{N}$ and p_k is well-defined due to $\overline{S} \cap \overline{B(0, \|t_k\|)}$ is compact, where $B(0, r) = \{x \mid \|x\| < r\}$. So we can get $0 \leq \|t_k - p_k\| \leq \|t_k\| \rightarrow 0$, then $0 \leq \|p_k\| \leq \|t_k\| + \|p_k - t_k\| \rightarrow 0$, that is $p_k \rightarrow 0$. Let $a_k = t_k - p_k, b_k = \frac{a_k}{\|a_k\|}, k \in \mathbb{N}$, then $\{b_k \mid k \in \mathbb{N}\}$ is a subset of unit ball. Due to unit ball is compact, unit ball is sequence compact, thus $\{b_k \mid k \in \mathbb{N}\}$ contains a converncible sub-sequence $\{b_{k_i}\}_{i \in \mathbb{N}}$, whose limit is contained in unit ball. Let $b_k \rightarrow b$, where $\|b\| = 1$.

From here, we will prove $b^T z \leq 0, \forall z \in \overline{S}$. $\forall z \in \overline{S}, k \in \mathbb{N}$, since $p_k + \lambda(z - p_k) \in \overline{S}, \forall \lambda \in [0, 1]$ by convexity of S and the definition of p_k , we can get $\phi(\lambda) = \|t_k - (p_k + \lambda(z - p_k))\|^2 \geq \phi(0) = \|t_k - p_k\|^2, \forall \lambda \in [0, 1]$. Thus, we can get $\phi'_+(0) = \lim_{\lambda \rightarrow 0^+} \frac{\phi(\lambda) - \phi(0)}{\lambda} \geq 0$. Since $\phi'(\lambda) = -2(t_k - p_k - \lambda(z - p_k))^T(z - p_k)$, then $\phi'(\lambda)$ is continuous. So $\phi'(0) = \phi'_+(0) \geq 0$. That is $-2a_k^T(z - p_k) \geq 0$, so $a_k^T(z - p_k) \leq 0$, then $b_k^T(z - p_k) \leq 0$. So $b_k^T z \leq b_k^T p_k \leq \|b_k\| \|p_k\| = \|p_k\|$. Then $b_{k_j}^T z \leq \|p_{k_j}\|$. Since $b_{k_j}^T z \rightarrow b^T z, \|p_{k_j}\| \rightarrow 0$, we can get $b^T z \leq 0$. Therefore $\{b^T z \leq 0 \mid z \in S\}$.

□

PROBLEM V Let K^* be the dual cone of a convex cone K . Prove the following.

1. K^* is a convex cone.
2. $K_1 \subset K_2$ implies $K_2^* \subset K_1^*$.
3. K^* is closed.
4. $\text{int} K^* = \{y \mid y^T x > 0, \forall x \in K\}$.
5. If $\text{int} K \neq \emptyset$, then K^* is pointed.
6. K^{**} is the closure of K .
7. If the closure of K is pointed then K^* has nonempty interior.

SOLUTION. 1. Since $K^* = \{y \mid y^T x \geq 0, \forall x \in K\}$, we can get $\forall y \in K^*, \forall \theta \geq 0, (\theta y)^T x = \theta(y^T x) \geq 0$, so $\theta y \in K^*$, thus, K^* is cone. $\forall y_1, y_2 \in K^*, \forall \lambda \in [0, 1], (\lambda y_1 + (1 - \lambda)y_2)^T x = \lambda y_1^T x + (1 - \lambda)y_2^T x \geq 0$. So $\lambda y_1 + (1 - \lambda)y_2 \in K^*$, thys K^* is convex.

2. $\forall y \in K_2^*$, then $\forall x \in K_2, y^T x \geq 0$. Since $K_1 \subset K_2$, then $\forall x \in K_1, y^T x \geq 0$, so $y \in K_1^*$. Thus, $K_1^* \supset K_2^*$.

3. Let $\phi(y) = y^T x$, then $\phi(y)$ is continuous on $\mathbb{R}^n$. So $\forall y_n \in K^*$ satisfies $y_n \rightarrow y, n \rightarrow \infty$, then $\phi(y_n) \rightarrow \phi(y)$. Since $y_n \in K^*$, then $\phi(y_n) = y_n^T x \geq 0$, then $y^T x = \phi(y) = \lim \phi(y_n) \geq 0$. Thus, $y \in K^*$. Therefore K^* is closed.

4. There is some thing wrong with the problem. We can prove that when K is closed and $0 \notin K$, $\text{int} K^* = \{y \mid y^T x > 0, \forall x \in K\}$. Otherwise, there are counterexample. If $0 \in K$, then $y^T 0 = 0 \forall y \in \mathbb{R}$, then $\{y \mid y^T x > 0, \forall x \in K\} = \emptyset$. While $K = \{(x, 0) \mid x \geq 0\}$, $K^* = \{(x, y) \mid x \geq 0\}$, so $\text{int} K^* = \{(x, y) \mid x > 0\} \neq \emptyset$. If K is not closed, consider

$K = \{(x, y) \mid x > 0, y > 0\}$, then $ux + vy \geq 0, \forall x, y > 0$, then $u, v \geq 0$. So $K^* = \{(u, v) \mid u \geq 0, v \geq 0\}$. Then $(1, 0) \notin \text{int}K^*$, but $(1, 0)^T(x, y) = x > 0$.

Next, we will prove $\text{int}K^* = \{y \mid y^T x > 0, \forall x \in K\}$, where K is closed and $0 \notin K$.

$\text{int}K^* \supset \{y \mid y^T x > 0, \forall x \in K\}$: Since K is closed and $0 \notin K$, then $\{\frac{x}{\|x\|} \mid x \in K\}$ is closed and well-defined. Let $X = \{\frac{x}{\|x\|} \mid x \in K\}$, then X is closed and bounded in $\mathbb{R}^n$. Therefore, X is compact. $\forall y \in \mathbb{R}$ satisfies $y^T x > 0, \forall x \in K$, then $y^T \frac{x}{\|x\|} > 0, \forall x \in K$. Thus, $y^T u > 0, \forall u \in X$. By compactness of X , we can get $\exists u_0 \in X, \inf_{u \in X} y^T u = y^T u_0 > 0$. Let $\varepsilon = \frac{1}{y^T u_0}$, $\forall e : \|e\| = 1, (y - \varepsilon e)^T u = y^T u - \varepsilon e^T u \geq y^T u - \varepsilon \geq \frac{1}{2}\varepsilon > 0$. So $\forall z \in B(y, \varepsilon) = \{w \mid \|w - y\| < \varepsilon\}$, $z^T u > 0 \forall u \in X$, then $\forall x \in K, z^T(\frac{x}{\|x\|}) > 0$, thus, $z^T x > 0$. Therefore $y \in \text{int}K^*$.

$\text{int}K^* \subset \{y \mid y^T x > 0, \forall x \in K\}$: $\forall y \in \text{int}K^*, \exists \varepsilon > 0, \forall e : \|e\| = 1$, we can get $y + \varepsilon e \in K^*$. $\forall x \in K, x \neq 0, \frac{x}{\|x\|} \in K$, then $y - \varepsilon x \in K^*$. So $(y - \varepsilon \frac{x}{\|x\|})^T \frac{x}{\|x\|} = y^T \frac{x}{\|x\|} - \varepsilon \geq 0$, so $y^T \frac{x}{\|x\|} \geq \varepsilon > 0$. Therefore $y \in \{y \mid y^T x > 0, \forall x \in K\}$.

5. Since $\text{int}K \neq \emptyset$, let $x \in \text{int}K$. Then $\exists \varepsilon > 0, \forall e : \|e\| = 1, x + \varepsilon e \in K$. If $\exists y \in K^* \cap (-K^*)$ and $y \neq 0$, then $-y \in K^*$, then $y^T(x + \varepsilon e) \geq 0$ and $(-y)^T(x - \varepsilon x) \geq 0$. Add these two inequation, we can get $2\varepsilon y^T e \geq 0, \forall e : \|e\| = 1$. Let $e = \frac{-y}{\|y\|}$, we can get $2\varepsilon(-\|y\|) \geq 0$. Contradiction! Therefore, $K^* \cap \{-K^*\} = \{0\}$.
6. $\forall y \in K^*, y$ satisfies $\forall x \in K, y^T x \geq 0$, then $x^T y = y^T x \geq 0$. Therefore $x \in K^{**}$. So $K \subset K^{**}$. Since K^{**} is closed, then $\overline{K} \subset K^{**}$. Next we prove $K^{**} \subset \overline{K}$. If $K^{**} \setminus \overline{K} \neq \emptyset$. Let $u \in K^{**} \setminus \overline{K}$, then $\exists \delta > 0$, we can get $0 < \delta < \|u - x\|, \forall x \in \overline{K}$. So by strict convex separating theorem, $\exists a, b$ satisfies $a^T x > b, \forall x \in \overline{K}$ and $a^T u < b$. Since K is cone, then $\theta a^T x > b, \forall \theta \geq 0$, then $b \leq 0$, so $a^T u < 0$. If $a^T x < 0, \exists x \in K$. Then $\exists \theta$ big enough satisfies $\theta a^T u x < b$. Therefore, $a^T x \geq 0, \forall x \in K$. Then $a \in K^*$. Since $u \in K^{**}$, then $\forall y \in K^*, u^T y \geq 0$. Then $a^T u = u^T a < 0$, while $a \in K^*$. Contradiction!
7. Suppose that $\overline{K}$ is pointed. We show that $\text{int}K^* \neq \emptyset$.

Let $C = \overline{K} \cap \{x \in \mathbb{R}^n \mid \|x\|_2 = 1\}$. Since $\overline{K}$ is closed, C is compact.

We claim that $0 \notin \text{conv}(C)$. Otherwise, there exist $x_1, \dots, x_m \in C$ and $\lambda_i \geq 0, \sum_{i=1}^m \lambda_i = 1$, such that $\sum_{i=1}^m \lambda_i x_i = 0$. Choose j such that $\lambda_j > 0$. Then $x_j = -\frac{1}{\lambda_j} \sum_{i \neq j} \lambda_i x_i$. Since $x_i \in \overline{K}$ and $\overline{K}$ is a convex cone, $\frac{1}{\lambda_j} \sum_{i \neq j} \lambda_i x_i \in \overline{K}$. Hence $x_j \in \overline{K} \cap (-\overline{K})$. But $\overline{K}$ is pointed, so $\overline{K} \cap (-\overline{K}) = \{0\}$, which contradicts $\|x_j\|_2 = 1$. Therefore $0 \notin \text{conv}(C)$.

Since $\text{conv}(C)$ is compact and convex, it can be strictly separated from the origin. Thus there exist $y \neq 0$ and $\alpha > 0$ such that $y^T x \geq \alpha > 0, \forall x \in C$.

Now let $x \in \overline{K} \setminus \{0\}$. Since $\overline{K}$ is a cone, $\frac{x}{\|x\|_2} \in C$. Therefore $y^T x = \|x\|_2 y^T \frac{x}{\|x\|_2} \geq \alpha \|x\|_2 > 0$. Hence $y^T x > 0, \forall x \in \overline{K} \setminus \{0\}$.

Since $K^* = (\overline{K})^*$, by the characterization of the interior of the dual cone,

$$\text{int}K^* = \{y \mid y^T x > 0, \forall x \in \overline{K} \setminus \{0\}\},$$

we conclude that $y \in \text{int}K^*$. Therefore $\text{int}K^* \neq \emptyset$.

□

PROBLEM VI Define monotone nonnegative cone as

$$K_{m+} = \{x \in \mathbb{R}^n \mid x_1 \geq x_2 \geq \cdots \geq x_n \geq 0\}.$$

1. Show K_{m+} is a proper cone.
2. Find K_{m+}^* .

SOLUTION. 1. Recall that a proper cone is a closed, convex, pointed cone with nonempty interior.

First, we will prove K_{m+} is a cone. If $x \in K_{m+}$ and $\theta \geq 0$, then

$$\theta x_1 \geq \theta x_2 \geq \cdots \geq \theta x_n \geq 0,$$

so $\theta x \in K_{m+}$.

Obviously, $K_{m+} = \{x \in \mathbb{R}^n \mid x_i - x_{i+1} \geq 0, i = 1, \dots, n-1, x_n \geq 0\}$. Thus, K_{m+} is an intersection of closed halfspaces, so it is closed and convex.

Next, we will prove it is pointed, suppose $x \in K_{m+}$, $-x \in K_{m+}$. Since $x_i \geq 0$ and $-x_i \geq 0$, we have $x_i = 0$. we obtain $x = 0$. Thus $K_{m+} \cap (-K_{m+}) = \{0\}$, so K_{m+} is pointed.

Finally, $(n, n-1, \dots, 1)$ satisfies all the defining inequalities strictly. Hence $\text{int } K_{m+} \neq \emptyset$. Therefore K_{m+} is a proper cone.

2. Define $z_i = x_i - x_{i+1}$, $i = 1, \dots, n-1$, $z_n = x_n$. Then $x \in K_{m+}$ if and only if $z_i \geq 0$, $i = 1, \dots, n$, and

$$x_i = \sum_{j=i}^n z_j.$$

Hence

$$x = z_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + z_2 \begin{pmatrix} 1 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + \cdots + z_n \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}.$$

Therefore K_{m+} is generated by $q_k = (\underbrace{1, \dots, 1}_k, 0, \dots, 0)^T$, $k = 1, \dots, n$.

By definition, $y \in K_{m+}^* \iff y^T x \geq 0$ for all $x \in K_{m+}$. Since every $x \in K_{m+}$ is a nonnegative linear combination of $q_1, \dots, q_n$, this is equivalent to $y^T q_k \geq 0$, $k = 1, \dots, n$. But $y^T q_k = \sum_{i=1}^k y_i$. Thus $K_{m+}^* = \left\{ y \in \mathbb{R}^n \mid \sum_{i=1}^k y_i \geq 0, k = 1, \dots, n \right\}$.

□